

Course Code : ECT 359-3

WYXZ/MS – 23 / 1605

**Sixth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

BIOMEDICAL ELECTRONICS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) Draw neat and clean diagram, wherever necessary.
- (3) Assume suitable data wherever necessary.

1.
 - (a) Illustrate with the help of a block diagram a generalized medical instrumentation system with supporting example. 5(CO1,2)
 - (b) Explain the origin of electrical activity of the heart with the help of a diagram. Draw a typical ECG waveform and label it. 5(CO1)
2.
 - (a) What are the types of external pacemakers ? Explain with the help of a diagram the various shapes of impulses generated by different types of external pacemakers. 5(CO2)
 - (b) Describe the various types of pre-amplifiers used in medical instruments, giving their important application areas. What is a differential amplifier and why is it preferred for applications in recording of bioelectric signals ? 5(CO2)
3.
 - (a) Mention any 5 biomedical recorders with their applications in medical diagnosis. Explain the function of an 'Electromyograph' biomedical recorder. 5(CO1,2)
 - (b) What do you mean by bio-telemetry ? Which type of modulation system is used in a multichannel bio telemetry system ? Draw and briefly elaborate the single channel telemetry system suitable for transmission of an ECG. 5(CO1,4)

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4. (a) Draw the block diagram of a pulse height analyzer and explain its working. What is the function of a pulse-height analyzer. 5(CO3)
- (b) Define characteristic impedance in ultrasound. Why a coupling medium is essential when using ultrasound for diagnostic purposes ? What is the frequency range of ultrasound normally used in medical field for diagnostic applications ? What are the any two advantages of an ultrasound imaging system ? 5(CO3)
5. (a) Describe the objective of using 'Patient Monitoring System'. List out the important parameters which are essential for patient monitoring. 5(CO4)
- (b) What is the need of 'Ambulatory Monitoring Systems' ? Explain the sub-systems for data recording and analysis in an ambulatory monitoring system with an example. 5(CO4)
6. (i) Distinguish between gross shock and micro current shock. Briefly elaborate on effect of various levels of current on the human body.
- (ii) Explain the possible effects of electric current on the human body. 10(CO5)

